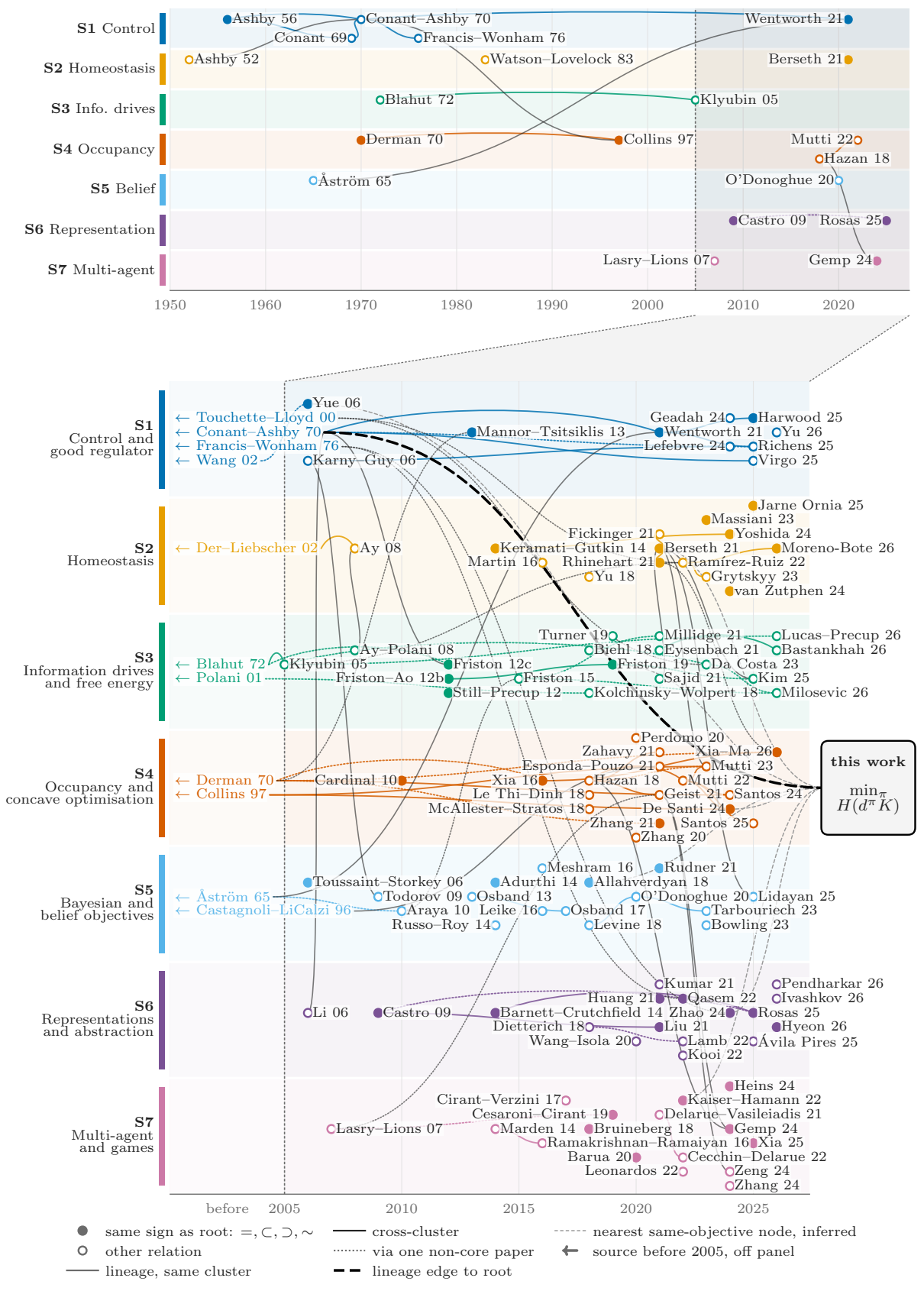

S1 Control
S2 Homeostasis
S3 Info. drives
S4 Occupancy
S5 Belief
S6 Representation
S7 Multi-agent
Ashby 56
Conant 69
Conant–Ashby 70
Francis–Wonham 76
Wentworth 21
Ashby 52
Watson–Lovelock 83
Berseth 21
Blahut 72
Klyubin 05
Derman 70
Collins 97
Mutti 22
Hazan 18
Åström 65
O'Donoghue 20
Castro 09
Rosas 25
Lasry–Lions 07
Gemp 24
1950
1960
1970
1980
1990
2000
2010
2020
S1
Control and good regulator
← Touchette–Lloyd 00
← Conant–Ashby 70
← Francis–Wonham 76
← Wang 02
Yue 06
Karny–Guy 06
Mannor–Tsitsiklis 13
Geadah 24
Harwood 25
Wentworth 21
Yu 26
Lefebvre 24
Richens 25
Virgo 25
S2
Homeostasis
← Der–Liebscher 02
Ay 08
Jarne Ornia 25
Massiani 23
Fickinger 21
Yoshida 24
Keramati–Gutkin 14
Berseth 21
Moreno-Bote 26
Martin 16
Rhinehart 21
Ramírez-Ruiz 22
Yu 18
Grytskyy 23
van Zutphen 24
S3
Information drives and free energy
← Blahut 72
← Polani 01
Klyubin 05
Ay–Polani 08
Friston–Ao 12b
Friston 12c
Still–Precup 12
Turner 19
Biehl 18
Friston 15
Friston 19
Millidge 21
Eysenbach 21
Sajid 21
Kolchinsky–Wolpert 18
Lucas–Precup 26
Bastankhah 26
Da Costa 23
Kim 25
Milosevic 26
S4
Occupancy and concave optimisation
← Derman 70
← Collins 97
Cardinal 10
Perdomo 20
Zahavy 21
Xia–Ma 26
Esponda–Pouzo 21
Mutti 23
Xia 16
Hazan 18
Mutti 22
Le Thi–Dinh 18
Geist 21
Santos 24
McAllester–Stratos 18
De Santi 24
Zhang 21
Santos 25
Zhang 20
this work
$\min_\pi H(d^\pi K)$
S5
Bayesian and belief objectives
← Åström 65
← Castagnoli–LiCalzi 96
Toussaint–Storkey 06
Todorov 09
Araya 10
Russo–Roy 14
Osband 13
Leike 16
Meshram 16
Rudner 21
Adurthi 14
Allahverdyan 18
O'Donoghue 20
Osband 17
Levine 18
Lidayan 25
Tarbouriech 23
Bowling 23
S6
Representations and abstraction
Li 06
Castro 09
Kumar 21
Pendharkar 26
Huang 21
Qasem 22
Ivashkov 26
Barnett–Crutchfield 14
Zhao 24
Rosas 25
Dietterich 18
Liu 21
Hyeon 26
Wang–Isola 20
Lamb 22
Ávila Pires 25
Kooi 22
S7
Multi-agent and games
Heins 24
Cirant–Verzini 17
Kaiser–Hamann 22
Cesaroni–Cirant 19
Delarue–Vasileiadis 21
Lasry–Lions 07
Marden 14
Bruineberg 18
Gemp 24
Ramakrishnan–Ramaiyan 16
Xia 25
Barua 20
Cecchin–Delarue 22
Leonardos 22
Zeng 24
Zhang 24
before 2005
2010
2015
2020
2025
same sign as root: =, ⊂, ⊃, ∼
other relation
lineage, same cluster
cross-cluster
via one non-core paper
lineage edge to root
nearest same-objective node, inferred
source before 2005, off panel